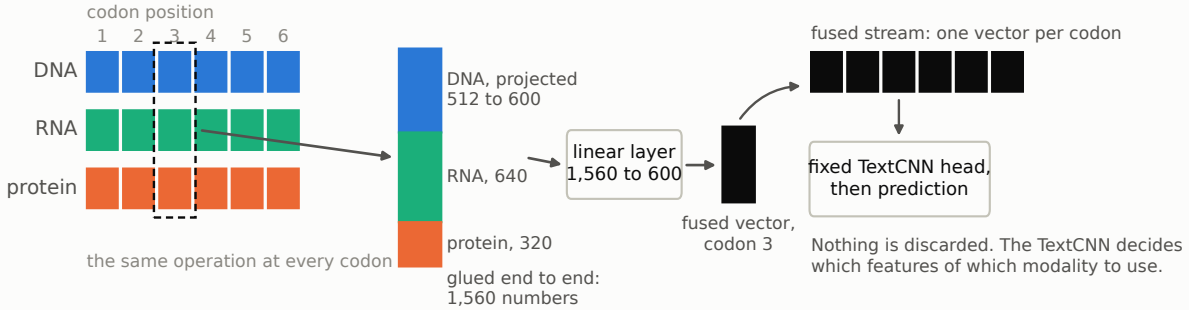
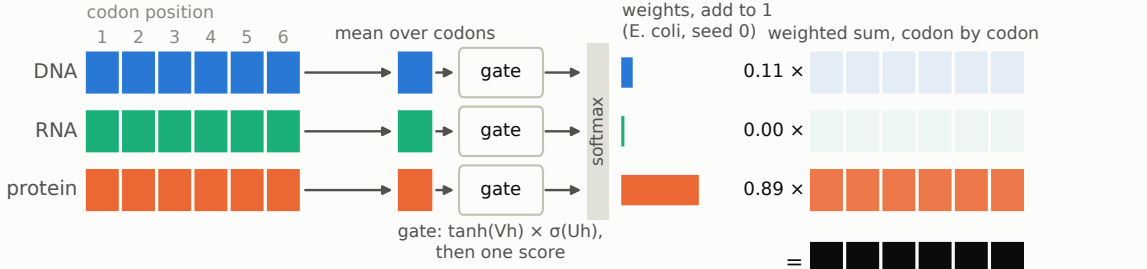
a Codon-aware concatenation: keep everything, position by position

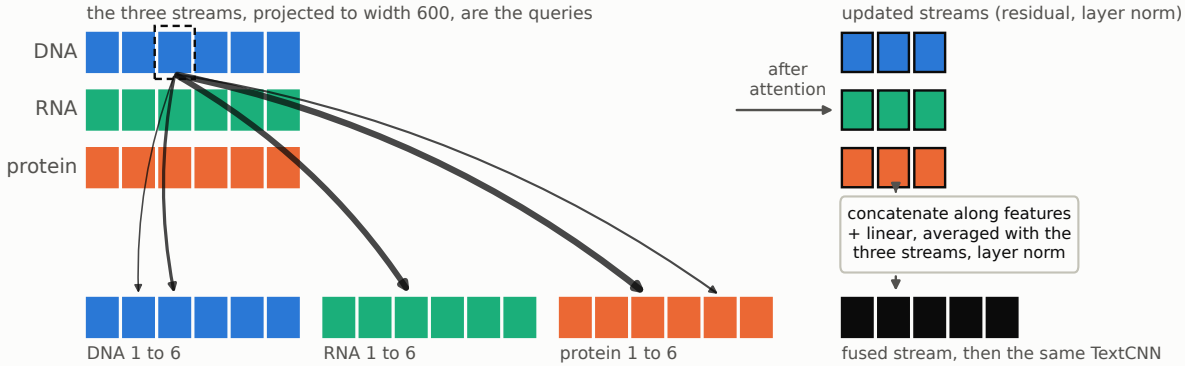


b Gated MIL attention: one weight per modality for the whole sequence



Entropy term in the loss: $\lambda \times \Sigma\ \alpha \log \alpha$ ($\lambda = 0.01$), which rewards sharp weights. The decision is three numbers per sequence: the head can choose a modality but cannot pick different features from different modalities at different codons. Here the RNA stream is multiplied by 0.00: dropped.

c Cross-modal attention: every position can read every modality at every position



context: the three streams laid end to end along the position axis, 18 positions (the keys and values).
DNA codon 3 as the query attends over all 18 positions; line width = attention weight (4 heads).
The RNA and protein streams do the same with their own attention; the three updated streams are then merged.

All three heads start from the same input (three streams on the codon grid, one vector per codon per modality) and end in the same fixed TextCNN (convolutions over 3, 4 and 5 codons, a max over positions, one linear layer).
Expressiveness: gated attention (three numbers per sequence) < concatenation (every feature, position by position) < cross-attention (every feature, any position to any position).